

AQA A-Level Further Mathematics (7367) Question Paper

Further Calculus

E1 - Improper Integrals

(A-Level)

Name: _____

Class: _____

Date: _____

Time: 218 min.

Marks: 184 marks

Comments:

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Q1.

A student states that $\int_0^{\frac{\pi}{2}} \frac{\cos x + \sin x}{\cos x - \sin x} dx$ is not an improper integral because $\frac{\cos x + \sin x}{\cos x - \sin x}$ is defined at both $x = 0$ and $x = \frac{\pi}{2}$.

Assess the validity of the student's argument.

(Total 2 marks)

Q2.

Evaluate the improper integral $\int_0^{\infty} \frac{4x - 30}{(x^2 + 5)(3x + 2)} dx$, showing the limiting process used. Give your answer as a single term.

(Total 8 marks)

Q3.

Show that the improper integral $\int_{25}^{\infty} \frac{1}{x\sqrt{x}} dx$ has a finite value and find that value.

(Total 4 marks)

Q4.

(a) Explain why $\int_0^1 x^4 \ln x dx$ is an improper integral. (1)

(b) Evaluate $\int_0^1 x^4 \ln x dx$ showing the limiting process used. (6)

(Total 7 marks)

Q5.

For the improper integral $\int_1^{\infty} x - 4(2x^2 - 5x) dx$, either show that the integral has a finite value and state its value, or explain why the integral does not have a finite value.

(Total 3 marks)

Q6.

Evaluate the improper integral

$$\int_0^{\infty} \left(\frac{2x}{x^2 + 4} - \frac{4}{2x + 3} \right) dx$$

showing the limiting process used and giving your answer in the form $\ln k$, where k is a constant.

(Total 6 marks)

Q7.

Show that only one of the following improper integrals has a finite value, and find that value:

(a) $\int_8^{\infty} x^{-\frac{2}{3}} dx;$

(b) $\int_8^{\infty} x^{-\frac{4}{3}} dx.$

(Total 5 marks)

Q8.

(a) Explain why $\int_{\frac{1}{2}}^{\infty} \frac{x(1-2x)}{x^2 + 3e^{4x}} dx$ is an improper integral.

(1)

(b) By using the substitution $u = x^2 e^{-4x} + 3$, find

$$\int \frac{x(1-2x)}{x^2 + 3e^{4x}} dx$$

(3)

(c) Hence evaluate $\int_{\frac{1}{2}}^{\infty} \frac{x(1-2x)}{x^2 + 3e^{4x}} dx$, showing the limiting process used.

(4)

(Total 8 marks)

Q9.

(a) Find $\int x^2 e^{-x} dx$.

(4)

(b) Hence evaluate $\int_0^{\infty} x^2 e^{-x} dx$, showing the limiting process used.

(3)

(Total 7 marks)

Q10.

(a) Find, in terms of p and q , the value of the integral $\int_p^q \frac{2}{x^3} dx$.

(3)

(b) Show that only one of the following improper integrals has a finite value, and find

that value:

(i) $\int_0^2 \frac{2}{x^3} dx$,

(ii) $\int_0^{\infty} \frac{2}{x^3} dx$.

(3)
(Total 6 marks)

Q11.

(a) Write $\frac{4}{4x+1} - \frac{3}{3x+2}$ in the form $\frac{C}{(4x+1)(3x+2)}$, where C is a constant.

(1)

(b) Evaluate the improper integral

$$\int_1^{\infty} \frac{10}{(4x+1)(3x+2)} dx$$

showing the limiting process used and giving your answer in the form $\ln k$, where k is a constant.

(6)
(Total 7 marks)

Q12.

(a) Find $\int x^2 \ln x dx$.

(3)

(b) Explain why $\int_0^e x^2 \ln x dx$ is an improper integral.

(1)

(c) Evaluate $\int_0^e x^2 \ln x dx$, showing the limiting process used.

(3)
(Total 7 marks)

Q13.

(a) Explain why $\int_0^{\frac{1}{16}} x^{-\frac{1}{2}} dx$ is an improper integral.

(1)

(b) For each of the following improper integrals, find the value of the integral or explain briefly why it does not have a value:

(i) $\int_0^{\frac{1}{16}} x^{-\frac{1}{2}} dx;$

(3)

(ii) $\int_0^{\frac{1}{16}} x^{-\frac{5}{4}} dx.$

(3)

(Total 7 marks)

Q14.

(a) Explain why $\int_1^{\infty} \frac{\ln x^2}{x^3} dx$ is an improper integral.

(1)

(b) (i) Show that the substitution $\frac{1}{x} = y$ transforms $\int \frac{\ln x^2}{x^3} dx$ into $\int 2y \ln y dy$.

(2)

(ii) Evaluate $\int_0^1 2y \ln y dy$, showing the limiting process used.

(5)

(iii) Hence write down the value of $\int_1^{\infty} \frac{\ln x^2}{x^3} dx$.

(1)

(Total 9 marks)

Q15.

(a) Explain why $\int_1^{\infty} 4xe^{-4x} dx$ is an improper integral.

(1)

(b) Find $\int 4xe^{-4x} dx$.

(3)

(c) Hence evaluate $\int_1^{\infty} 4xe^{-4x} dx$, showing the limiting process used.

(3)

(Total 7 marks)

Q16.

For each of the following improper integrals, find the value of the integral or explain why it does not have a value:

(a) $\int_1^{\infty} x^{-\frac{1}{4}} dx;$

(3)

(b) $\int_1^{\infty} x^{-\frac{5}{4}} dx$;

(3)

(c) $\int_1^{\infty} (x^{-\frac{3}{4}} - x^{-\frac{5}{4}}) dx$.

(1)

(Total 7 marks)

Q17.

- (a) Use integration by parts to show that $\int \ln x \, dx = x \ln x - x + c$, where c is an arbitrary constant.

(2)

- (b) Hence evaluate $\int_0^1 \ln x \, dx$, showing the limiting process used.

(4)

(Total 6 marks)

Q18.

Evaluate the improper integral

$$\int_1^{\infty} \left(\frac{1}{x} - \frac{4}{4x+1} \right) dx$$

showing the limiting process used and giving your answer in the form $\ln k$, where k is a constant to be found.

(Total 5 marks)

Q19.

- (a) Explain why $\int_1^{\infty} xe^{-3x} \, dx$ is an improper integral.

(1)

- (b) Find $\int xe^{-3x} \, dx$.

(3)

- (c) Hence evaluate $\int_1^{\infty} xe^{-3x} \, dx$, showing the limiting process used.

(3)

(Total 7 marks)

Q20.

For each of the following improper integrals, find the value of the integral or explain briefly why it does not have a value:

(a) $\int_0^{\infty} \frac{1}{\sqrt{x}} dx$; (3)

(b) $\int_0^{\infty} \frac{1}{x\sqrt{x}} dx$. (4)

(Total 7 marks)

Q21.

(a) Find $\int x^3 \ln x dx$. (3)

(b) Explain why $\int_0^e x^3 \ln x dx$ is an improper integral. (1)

(c) Evaluate $\int_0^e x^3 \ln x dx$, showing the limiting process used. (3)

(Total 7 marks)

Q22.

(a) Explain why $\int_0^e \frac{\ln x}{\sqrt{x}} dx$ is an improper integral. (1)

(b) Use integration by parts to find $\int x^{-\frac{1}{2}} \ln x dx$. (3)

(c) Show that $\int_0^e \frac{\ln x}{\sqrt{x}} dx$ exists and find its value. (4)

(Total 8 marks)

Q23.

For each of the following improper integrals, find the value of the integral or explain briefly why it does not have a value:

(a) $\int_0^1 (x^{\frac{1}{3}} + x^{-\frac{1}{3}}) dx$ (4)

(b) $\int_0^1 \frac{x^{\frac{1}{3}} + x^{-\frac{1}{3}}}{x} dx.$

(4)

(Total 8 marks)

Q24.

- (a) Write down the value of

$$\lim_{x \rightarrow \infty} x e^{-x}$$

(1)

- (b) Use the substitution $u = x e^{-x} + 1$ to find $\int \frac{e^{-x}(1-x)}{x e^{-x} + 1} dx.$

(2)

- (c) Hence evaluate $\int_1^{\infty} \frac{1-x}{x+e^x} dx,$ showing the limiting process used.

(4)

(Total 7 marks)

Q25.

- (a) For each of the following improper integrals, find the value of the integral or explain briefly why it does not have a value:

(i) $\int_0^9 \frac{1}{\sqrt{x}} dx;$

(3)

(ii) $\int_0^9 \frac{1}{x\sqrt{x}} dx.$

(3)

- (b) Explain briefly why the integrals in part (a) are improper integrals.

(1)

(Total 7 marks)

Q26.

- (a) Find $\int_0^a x e^{-2x} dx,$ where $a > 0.$

(5)

- (b) Write down the value of $\lim_{a \rightarrow \infty} a^k e^{-2a},$ where k is a positive constant.

(1)



- (c) Hence find $\int_0^{\infty} x e^{-2x} dx$.

(2)

(Total 8 marks)

Q27.

- (a) Show that $\lim_{a \rightarrow \infty} \left(\frac{3a+2}{2a+3} \right) = \frac{3}{2}$

(2)

- (b) Evaluate $\int_1^{\infty} \left(\frac{3}{3x+2} - \frac{2}{2x+3} \right) dx$, giving your answer in the form $\ln k$, when k is a rational number.

(5)

(Total 7 marks)

Q28.

For each of the following improper integrals, find the value of the integral or explain briefly why it does not have a value:

- (a) $\int_2^{\infty} 8x^{-3} dx$

(3)

- (b) $\int_2^{\infty} (8x^{-3} + 1) dx$

(1)

- (c) $\int_2^{\infty} 8x^{-3}(x+1) dx$

(3)

(Total 7 marks)